



ST. ANDREW'S JUNIOR COLLEGE
Higher 1
PRELIMINARY EXAMINATION 2012

Name _____

Class _____

CHEMISTRY

PAPER 2

Candidates are to answer Section **A** on the Question Paper.

Additional Materials: Writing paper
 Data Booklet

8872/ 02

10 September 2012

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **two** questions on separate writing paper.

At the end of the exam, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For examiner's use	
Section A	/ 40
B5	
B6	
B7	
Total	/ 80

This paper consists of **16** printed pages **including** this page.

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Section A

Answer **all** questions in the spaces provided.

1. Magnesium exists as three principal isotopes, ^{24}Mg , ^{25}Mg and ^{26}Mg .

- (a) Use the following data to calculate the relative atomic mass of magnesium to **two** decimal places.

Isotope	Relative isotopic mass	Natural abundance (%)
^{24}Mg	23.99	78.99
^{25}Mg	24.99	10.00
^{26}Mg	25.98	11.01

[1]

- (b) Magnesium and phosphorus are both in period 3. Describe and explain the difference in the size of their atomic radii.

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[2]

[Turn Over

1. (c) Magnesium and phosphorus form compounds with chlorine. The chlorides of the two elements

have different pH in water as shown in the table.

Compound in water	pH
MgCl ₂	6.5
PCl ₃	2

With the aid of balanced equations, describe and explain the differences in pH values.

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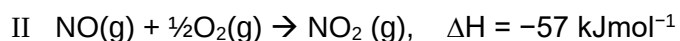
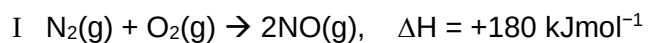
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[3]

[Total: 6 m]

2. One of the gases responsible for acid rain, nitrogen dioxide, NO₂, could be produced in the car engine by the following reaction sequence.



- (a) Draw a dot-and-cross diagram of a molecule of NO₂. Hence, state its shape and bond angle.

[2]

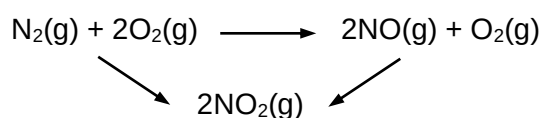
2. (b) Explain briefly why reaction I occurs in car engines despite the high activation energy, and

how NO gas can be removed in the car exhaust by a catalytic converter.

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[2]

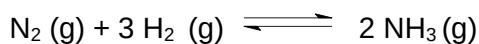
- (c) Using the following energy cycle or otherwise, calculate the enthalpy change of formation of NO₂.



[2]

[Total: 6 m]

3. The Haber process, is the nitrogen fixation reaction of nitrogen gas and hydrogen gas, over an enriched iron or ruthenium catalyst, which is used to industrially produce ammonia. The reaction is as follows:



- (a) (i) Write an expression for the equilibrium constant, K_c for the reaction.

[Turn Over

- 3(a) (ii) A mixture of N₂ and H₂ gases is heated in a 3.0 dm³ closed container and the reaction allowed

to reach equilibrium at 200 atm and 280 °C. The equilibrium mixture contains 0.80 mol of N_2 , 0.50 mol of H_2 and 1.60 mol of NH_3 . Calculate the value of K_c at 200 atm and 280 °C, stating the appropriate units.

- (iii) State and explain how the position of equilibrium and the equilibrium constant would change if the pressure is increased to 400 atm.

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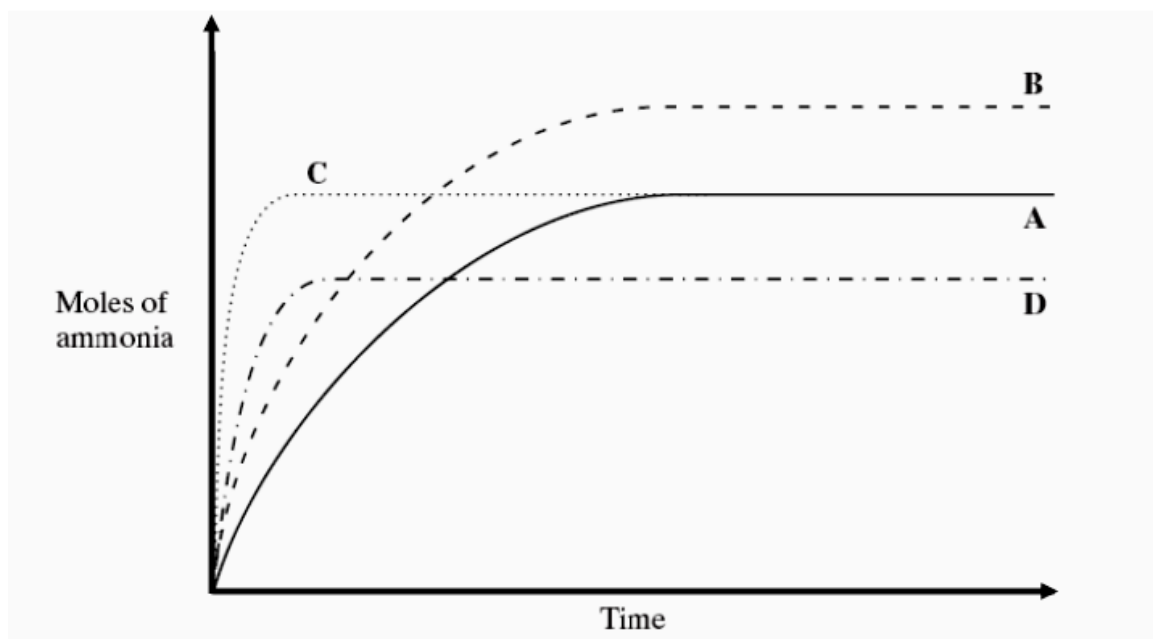
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[4]

3(b)

Experiment **A** was carried out starting with 2 mol of nitrogen and 6 mol of hydrogen at 200 atm and 280 °C. Curve **A** shows how the number of moles of ammonia changed with time in experiment **A**.

Curves **B**, **C** and **D** refer to similar experiments, starting with the same amount of reactants but each with different conditions.



- (i) Use Le Chatelier's principle to identify which one of the curves **B**, **C** or **D** represents an experiment carried out at the same temperature as experiment **A** but at a higher pressure. Explain your answer.

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- (ii) Suggest the change in conditions that would produce curve **C**?

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[3]

[Turn Over

- 3(c) (i)** Ammonia burns in oxygen to give nitrogen dioxide and steam. Write the equation which represents the enthalpy change of combustion of ammonia.

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- (ii)** Hence, use the following information to calculate a value for the enthalpy change of combustion of ammonia.

Enthalpy change of formation of nitrogen dioxide	-34 kJmol ⁻¹
Enthalpy change of formation of steam	-241.83 kJmol ⁻¹
Enthalpy change of formation of ammonia	-46 kJmol ⁻¹

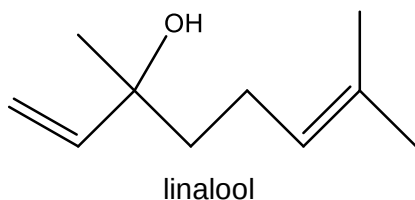
- (iii)** Using relevant information from the *Data Booklet* as well as your answer in **c(ii)**, calculate a value for the bond energy of the bond between the nitrogen atom and the oxygen atom in nitrogen dioxide, assuming that the bond energy of both bonds are the same.

- 3(c) (iv)** The combustion of 44 g of ammonia produces 89 g of nitrogen dioxide. Calculate the percentage efficiency of this reaction.

[6]

[Total: 13 m]

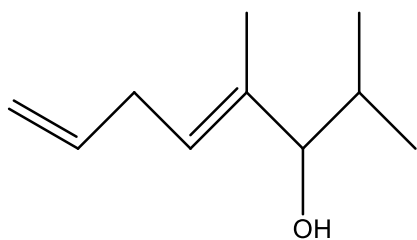
- 4. (a)** Lavender oil has long been used in aromatherapy. The scent has a calming effect which may aid in relaxation and the reduction of anxiety. Linalool is one of the primary components of lavender oil.



- (i)** Linalool reacts with Na metal to give compound **Y**. Explain with the aid of a diagram why compound **Y** is soluble in water.
- (ii)** Circle on the structure of linalool all the carbon atoms that are sp^3 hybridised.

[Turn Over

4(a) (iii) Suggest a simple chemical test to distinguish between linalool and its isomer, **Z**.



Z

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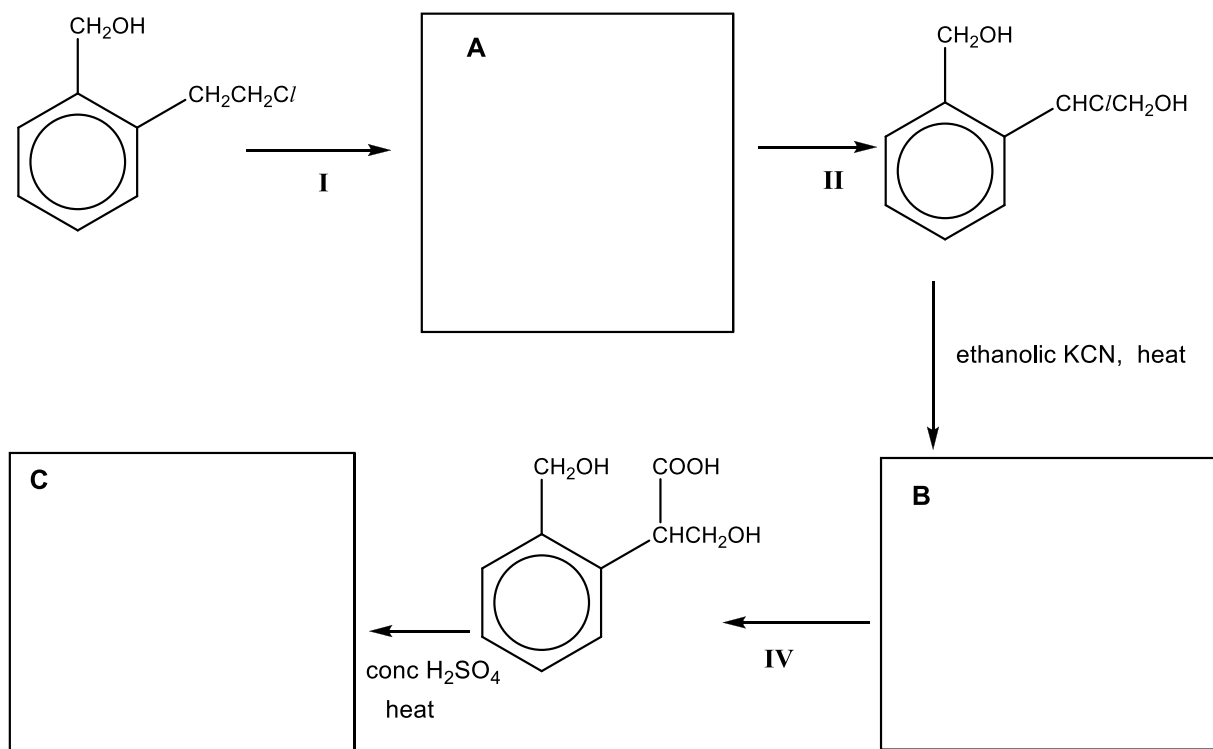
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(iv) Draw the structures of the organic products that are formed when hot acidified KMnO_4 is added to linalool.

[7]

4(b) The reaction scheme below shows the synthesis of a sweet-smelling compound **C**.



- (i) Draw the structures of **A** to **C** in the spaces provided
(ii) Give the reagents and conditions for steps **I**, **II** and **IV**.

I :
II :
IV :

- (iii) Name the type of reaction for steps **II** and **IV**

II :
IV :

[8]

[Total: 15 m]

[Turn Over

Section B

Answer 2 out of 3 questions.

5. Ethanedioic acid, $(\text{COOH})_2$, is often used a cleaning agent for the removal of rust.

(a) Draw a dot-and-cross diagram to show the arrangement of electrons in the $(\text{COOH})_2$ molecule. Hence, suggest the bond angle about either carbon atom.

[2]

(b) Ethanedioic acid is a *weak Bronsted acid*.

What do you understand by the term in italics?

[2]

(c) A solution containing $(\text{COOH})_2$ and its salt, $\text{HOOC-COO}^- \text{Na}^+$, acts as a buffer solution.

(i) What do you understand by the term *buffer solution*?

(ii) Write ionic equations to show how this solution reacts with a small amount of

I H^+ (aq) ions,

II OH^- (aq) ions.

[3]

(d) Ethanedioic acid and acidified potassium manganate(VII) react together to form carbon dioxide.

When a 0.32 g impure sample of ethanedioic acid was reacted with an excess of acidified potassium manganate(VII), the carbon dioxide gas produced was bubbled into an excess of $\text{Ba}(\text{OH})_2$ solution. The white precipitate formed was dried and was found to weigh 0.788 g.

5. (d) (i) Write an ion-electron equation for the formation of CO_2 from ethanedioic

acid.

Hence write a balanced equation for the reaction between ethanedioic acid and acidified potassium manganate(VII).

- (ii) State the identity of the white precipitate formed from the reaction of CO_2 and $\text{Ba}(\text{OH})_2$.
- (iii) Calculate the percentage purity of ethanedioic acid in the sample.

[7]

- (e) Give the reagents, conditions and the structure of all intermediates to show how you would carry out the synthesis of ethanedioic acid from

- (i) $\text{CH}_2=\text{CH}_2$
- (ii) $\text{CH}_3\text{COCO}_2\text{H}$

[6]

[Total: 20]

6. (a) The rate of reaction between 2-bromo-2-methylpropane, $(\text{CH}_3)_3\text{CBr}$, and aqueous sodium hydroxide when heated under reflux is found to be independent of $[\text{OH}^-]$ but directly proportional to $[(\text{CH}_3)_3\text{CBr}]$. Hence the rate equation is

$$\text{Rate} = k[(\text{CH}_3)_3\text{CBr}]$$

- (i) State the type of reaction above.
- (ii) Write a balanced chemical equation for the reaction.
- (iii) Sketch a graph of rate against concentration for
 - I $(\text{CH}_3)_3\text{CBr}$
 - II OH^-
- (iv) Predict and explain the relative rate of reaction if $(\text{CH}_3)_3\text{CCl}$ was used instead in the reaction above.

[Turn Over]

- 6 (a) (v) With the aid of an appropriate sketch of the energy distribution curve, explain why an increase in temperature can speed up this chemical

reaction.

[9]

- (b) Straight chain alkanes undergo thermal cracking to give shorter chain hydrocarbons. When straight chain $C_{14}H_{30}$ undergoes cracking, compounds **P**, **Q** and **R** are formed in the ratio of 2 : 1 : 1.

- (i) When 20 cm^3 of **P** is completely burnt in 150 cm^3 of oxygen gas, the resultant gas occupies 110 cm^3 and has an average relative molecular mass of 38.5.

Deduce the molecular formula of **P**.

(All volumes are measured at room temperature and pressure.)

- (ii) Using the following information, deduce the structure of **Q**.

- **Q** comprises 85.7% C and 14.3% H.
- **Q** exhibits geometric isomerism.
- 0.826 g of gaseous **Q** occupies 354 cm^3 at room temperature and pressure.

- (iii) Hence deduce the molecular formula of **R**.

- (iv) 1 mole of **R** reacts with 2 moles of aqueous Br_2 . Upon reaction with hot acidified $KMnO_4$, **R** produces only carbon dioxide and water.

Identify **R** and write equations for the reaction of **R** and aqueous Br_2 .

[11]

[Total: 20]

7. (a) Lemon juice contains citric acid which is a weak acid. The pH of lemon juice is

2.11.



The concentration of citric acid in lemon juice is determined by titrating 25.0 cm³ of the solution with 0.200 mol dm⁻³ of sodium hydroxide. The end-point was reached when 35.80 cm³ of sodium hydroxide was added.

- (i) Write an expression for the acid dissociation constant, K_a , of citric acid.
- (ii) Calculate the concentration of citric acid in lemon juice.
- (iii) Hence prove that citric acid is a weak acid.
- (iv) Given that $[\text{C}_6\text{H}_7\text{O}_7^-] = [\text{H}^+]$, calculate K_a of citric acid, stating its units.
- (iv) Write an equation to show that the salt formed between citric acid and sodium hydroxide is alkaline.
- (v) State a suitable indicator for the titration.

[8]

- (b) (i) Bauxite is an aluminium ore and is the main source of aluminium. To extract the aluminium in the ore, sodium hydroxide is used to dissolve aluminium oxide in the ore.

Give a balanced chemical equation for the reaction between sodium hydroxide and aluminium oxide.

- (ii) Another aluminium-containing compound is aluminium sulfate. A pure sample of aluminium sulfate was dissolved in water. When solid calcium carbonate was added into the solution, effervescence was observed.

Explain the reactions, giving equations where appropriate.

[4]

[Turn Over]

7. (c) Aluminium oxide can also be used in organic chemistry as a dehydrating agent.

When aluminium oxide reacts with compound **A**, $C_6H_{14}O$, two isomers **B** and **C** are formed. When the two isomers are reacted with hot acidified potassium manganate(VII), isomer **B** gives compound **D**, C_3H_6O , as the only product. While isomer **C** gives compound **E** and effervescence is seen. Both compounds **D** and **E** give an orange precipitate with 2,4-dinitrophenylhydrazine and a yellow precipitate in alkaline aqueous iodine.

Deduce the structures of compounds **A – E**. Explain the chemistry of the reactions.

[8]

[Total: 20]

End of Paper 2